DSC5001: Statistical Machine Learning I

Assignment 2

Semester A, 2026/27

Name: _____

Student ID: _____

Total: 100 marks **Deadline: November 5, 2026**

AI USE IS NOT PERMITTED FOR THIS ASSIGNMENT. All answers, calculations, and explanations must be your own work.

- There are six questions. Answer **all six**.
- Questions 1, 3, and 5 emphasize calculation and application (50 marks).
- Questions 2, 4, and 6 emphasize derivation and proof (50 marks).
- Questions 1–2 cover **Week 5** (nonlinear regression and splines); Questions 3–4 cover **Week 7** (local smoothing and Gaussian processes); Questions 5–6 cover **Week 8** (classification).
- This is a written assignment. No programming or coding submission is required.
- Show the essential calculations, derivations, and reasoning. Unsupported numerical answers may receive limited credit.
- Use methods from the lectures and tutorials. Any additional mathematical facts needed for a proof are stated in the relevant question. Use natural logarithms and give non-exact numerical results to at least three decimal places.
- Submit your own work. Handwritten solutions must be **scanned**, not photographed. Typed mathematics must use proper equation formatting.

Question	1	2	3	4	5	6	Total
Marks	15	15	15	20	20	15	100

Question 1: Polynomial Fits and Generalization [15 marks]

A regression study uses the five training observations

$$x = (-3, -1, 0, 1, 3), \qquad y = (9, 2, 1, 4, 11).$$

Consider a linear model and a quadratic model, both fitted by ordinary least squares (OLS):

$$M_1 : \ f_1(x) = a_0 + a_1 x, \qquad M_2 : \ f_2(x) = b_0 + b_1 x + b_2 x^2.$$

(a) Construct the two design matrices. For the quadratic model, compute $\boldsymbol{X}_2^\top \boldsymbol{X}_2$ and $\boldsymbol{X}_2^\top \boldsymbol{y}$. [3 marks]

(b) Obtain the OLS coefficient estimates for both models, showing the equations used to solve for the coefficients. [5 marks]

(c) Compute the training residual sum of squares (RSS) of both fits. Explain, using the relationship between the two model spaces, why quadratic OLS can never have *larger* training RSS than linear OLS on the same data. [3 marks]

(d) Two independent held-out observations are

$$(x_{\text{val}}, y_{\text{val}}) = (1.5, 4.5), \qquad (-1.5, 5.5).$$

Without refitting, calculate the *mean squared validation error* for each model. Select the model favored by these two observations and explain why this does not establish which model is universally better. [4 marks]

Question 2: A Cubic Spline with Two Knots [15 marks]

A curve is represented by a cubic regression spline with knots at $\xi_1 = 1$ and $\xi_2 = 3$:

$$s(x) = 1 + 2x + x^2 + \gamma_1(x-1)_+^3 + \gamma_2(x-3)_+^3, \qquad (u)_+ = \max(u, 0).$$

The coefficients γ_1 and γ_2 are unknown. Two measured values used to determine them are

$$s(2) = 11, \qquad s(4) = 78.$$

(a) Use the two values to solve for γ_1 and γ_2. Explain why the observation at $x = 2$ does not depend on γ_2. [4 marks]

(b) Write the resulting $s(x)$ as an explicit polynomial on each of the three intervals $x \leq 1$, $1 < x \leq 3$, and $x > 3$. [4 marks]

(c) Prove that s, s', and s'' are continuous at *both* knots. Compute the jump in the third derivative s''' from left to right at each knot. [5 marks]

(d) Explain which part of the input range changes if γ_2 is altered while γ_1 is fixed. Contrast this with changing γ_1. [2 marks]

Question 3: Kernel and Local Linear Regression [15 marks]

Five observations are available:

$$x = (0, 1, 2, 3, 4), \qquad y = (1, 3, 2, 6, 8).$$

Predict at $x_0 = 1$ using the triangular kernel with bandwidth $h = 3$:

$$K(u) = \max(1 - |u|, 0), \qquad w_i = K\left(\frac{x_i - x_0}{h}\right).$$

For local constant regression, let $\alpha_i = w_i / \sum_j w_j$. For local linear regression, fit $y_i \approx a + b(x_i - x_0)$ using these same weights.

(a) Compute the raw and normalized kernel weights, the Nadaraya–Watson prediction, and the effective sample size $n_{\text{eff}} = 1/\sum_i \alpha_i^2$. [5 marks]

(b) Write $u_i = x_i - x_0$. Form the 2×2 weighted normal equations for the local linear fit, solve for $(\widehat{a}, \widehat{b})$, and give the prediction at x_0. [5 marks]

(c) Omit the observation $(x, y) = (1, 3)$ without changing the bandwidth or kernel. Compute the leave-one-out *local constant* prediction at $x = 1$ and the squared prediction error. [2 marks]

(d) Now replace every response with $y_i' = y_i + 2(x_i - x_0)$. Without solving another set of normal equations, show why the *local linear prediction at* x_0 is unchanged. Calculate how much the *local constant prediction* changes. [3 marks]

Question 4: Deriving a Gaussian Process Update [20 marks]

Let f follow a zero-mean Gaussian process. We observe one noisy response

$$Y_1 = f(x_1) + \varepsilon_1, \qquad \varepsilon_1 \sim \mathcal{N}(0, \tau^2),$$

where ε_1 is independent of f. At a target x_*, write $F_* = f(x_*)$ and define

$$\mathrm{Var}(f(x_1)) = v_1 > 0, \qquad \mathrm{Var}(F_*) = v_* > 0, \qquad \mathrm{Cov}(f(x_1), F_*) = c.$$

Assume $\tau^2 \geq 0$, $v_1 + \tau^2 > 0$, and a valid joint Gaussian model. *You may use: linear transformations of jointly Gaussian variables are jointly Gaussian, and jointly Gaussian scalar variables with zero covariance are independent.*

(a) Derive the covariance matrix of $(Y_1, F_*)^\top$, explicitly accounting for observation noise. [3 marks]

(b) Define

$$A = \frac{c}{v_1 + \tau^2}, \qquad Z = F_* - AY_1.$$

Show that $\mathrm{Cov}(Z, Y_1) = 0$ and compute $\mathrm{Var}(Z)$. Justify the independence of Z and Y_1. [7 marks]

(c) Use $F_* = AY_1 + Z$ to derive the conditional mean and variance of F_* given $Y_1 = y_1$. Prove that the posterior variance cannot exceed v_* and explain the effect of increasing τ^2 while v_1, v_*, c stay fixed. [6 marks]

(d) For $v_1 = v_* = 1$, $c = 3/4$, $\tau^2 = 1/4$, and $y_1 = 2$, calculate the posterior mean and variance of F_* and the variance of a new response $Y_* = F_* + \varepsilon_*$, with independent $\varepsilon_* \sim \mathcal{N}(0, \tau^2)$. [4 marks]

Question 5: Gaussian Discriminant Decisions [20 marks]

A binary classifier uses a scalar measurement X and labels $Y \in \{0, 1\}$. The class priors and conditional distributions are

$$\pi_0 = \frac{1}{4}, \quad \pi_1 = \frac{3}{4}, \qquad X \mid Y = 0 \sim \mathcal{N}(1, 1), \quad X \mid Y = 1 \sim \mathcal{N}(4, 9).$$

For a one-dimensional Gaussian model, use the class score

$$\delta_k(x) = -\frac{1}{2} \log \sigma_k^2 - \frac{(x - \mu_k)^2}{2\sigma_k^2} + \log \pi_k,$$

where μ_k, σ_k^2, and π_k are class k's mean, variance, and prior. Assume equal misclassification costs unless stated otherwise.

(a) Derive $\Delta(x) = \delta_1(x) - \delta_0(x)$. Compute the posterior probabilities $\mathbb{P}(Y = 1 \mid X = x)$ at $x = 0$ and $x = 3$. [6 marks]

(b) Solve $\Delta(x) = 0$ for *all* QDA decision boundaries. Identify the intervals where the equal-cost classifier chooses each class. [5 marks]

(c) Replace the class-specific variances with one shared variance $\sigma^2 = 5$, keeping the same means and priors. Derive the *LDA* decision threshold under equal costs. [4 marks]

(d) Return to the original QDA model. Suppose a false positive costs 1, a false negative costs 4, and a correct decision costs 0. Derive the posterior-probability decision threshold, decide the class at $x = 0$, and explain why it differs from the equal-cost decision. [5 marks]

Question 6: Logistic Loss and Convexity [15 marks]

For binary labels $y_i \in \{0, 1\}$ and scalar inputs $x_i \in \mathbb{R}$, consider logistic regression

$$z_i = \beta_0 + \beta_1 x_i, \qquad p_i = \sigma(z_i) = \frac{1}{1 + e^{-z_i}},$$

with *summed* cross-entropy loss

$$\mathcal{L}(\beta_0, \beta_1) = -\sum_{i=1}^{n} \big[y_i \log p_i + (1 - y_i) \log(1 - p_i) \big].$$

For part (c), you may use the calculus criterion: a twice-differentiable function is convex if its Hessian (the matrix of second partial derivatives) is positive semidefinite, meaning $\boldsymbol{u}^\top \boldsymbol{H} \boldsymbol{u} \geq 0$ for every $\boldsymbol{u}$.

(a) Differentiate the sigmoid and show that $\sigma'(z) = \sigma(z)[1 - \sigma(z)]$. [2 marks]

(b) Starting from the given cross-entropy, derive $\partial \mathcal{L}/\partial \beta_0$ and $\partial \mathcal{L}/\partial \beta_1$. Show the chain-rule argument. [5 marks]

(c) Derive the 2×2 Hessian of $\mathcal{L}$. Prove it is positive semidefinite by computing $\boldsymbol{u}^\top \nabla^2 \mathcal{L}\, \boldsymbol{u}$ for an arbitrary $\boldsymbol{u} \in \mathbb{R}^2$, and state what this means for the training objective. [5 marks]

(d) Add an L_2 penalty only to the slope: $\mathcal{L}_\lambda = \mathcal{L} + \lambda \beta_1^2/2$, with $\lambda = 1$. For two observations $(x_1, y_1) = (-1, 0)$ and $(x_2, y_2) = (2, 1)$, start from $(\beta_0, \beta_1) = (0, 0)$. Compute one gradient-descent update of $\mathcal{L}_\lambda$ using learning rate $\rho = 1/4$. [3 marks]